

Contact Angle and Contact Radius Dynamics of a Drying Droplet: Evaporation Rates, Surface Tension Analysis, and Force Balance

Butle Vedika Ashirwad, Kumar Prince, Prakrut Moon, Prince Kumar, Priyanshu Patel, and Shah Mahi Sachinkumar

Abstract. We report a comprehensive experimental and theoretical study of evaporation dynamics of sessile droplets on hydrophilic (glass) and hydrophobic (oil-coated glass) substrates at 30 °C and 45 °C using smartphone side-view imaging. We derive and apply analytical expressions for the volumetric evaporation rate $|dV/dt|$, the instantaneous drying rate, and the full force balance at the three-phase contact line including solid–vapour, solid–liquid, and liquid–vapour surface tension components. For water droplets on glass, evaporation proceeds primarily in the Constant Contact Radius (CCR) mode with a nearly constant drying rate, while oil-coated substrates exhibit Constant Contact Angle (CCA) behaviour with a rapidly decaying rate. Temperature elevation from 30 °C to 45 °C accelerates evaporation by $\approx 40\%$ and shifts the CCR-to-CCA transition earlier. Ethanol droplets confirm the same qualitative behaviour with $\approx 2.5\times$ faster drying. The diffusion-limited evaporation framework, particularly the Popov geometric factor $f(\theta)$, is shown to be the dominant controller of evaporation rate.

Introduction

The evaporation of a sessile liquid droplet on a solid surface is a classical problem in interfacial fluid dynamics with relevance to inkjet printing, DNA micro-array fabrication, spray cooling, and colloidal self-assembly.⁽¹⁾ When a droplet sits on a surface, its shape at any instant is characterised by two observable parameters: the *contact radius* R , the radius of the liquid–solid contact circle, and the *contact angle* θ , the angle subtended at the three-phase contact line between the liquid–vapour and liquid–solid interfaces.

As evaporation proceeds, the droplet accommodates the loss of volume by adjusting R and θ . The manner in which it does so depends critically on surface wettability and the pinning strength of the contact line. Three canonical evaporation regimes have been identified:^(2; 3)

- **Constant Contact Radius (CCR):** The contact line is pinned; R remains fixed while θ decreases.
- **Constant Contact Angle (CCA):** The contact line recedes freely; θ stays near its equilibrium value while R decreases.
- **Stick–slip:** Intermittent pinning and depinning give rise to alternating CCR and CCA intervals.

The present work provides: (i) complete derivations of the instantaneous evaporation rate and drying rate in each mode; (ii) a full surface-tension force balance at the contact line; (iii) schematic diagrams of droplet geometry and force vectors; and (iv) experimental validation on two substrates at two temperatures for both water and ethanol. We further demonstrate that the Popov geometric factor $f(\theta)$ is the dominant controller

of evaporation rate, capable of producing $\sim 20\times$ rate differences from geometry alone.

Background Theory

Sessile droplet geometry

A droplet small enough that gravity is negligible compared with surface tension (Bond number $Bo = \rho g R^2 / \gamma < 1$) adopts the shape of a spherical cap. Its volume is

$$V = \frac{\pi h}{6} (3R^2 + h^2), \quad (1)$$

where $h = R \tan(\theta/2)$. Substituting:

$$V = \frac{\pi R^3}{3} \cdot \frac{(1 - \cos \theta)^2 (2 + \cos \theta)}{\sin^3 \theta} \equiv \frac{\pi R^3}{3} g(\theta). \quad (2)$$

The geometric factor $g(\theta)$ varies from 0 at $\theta = 0$ to $4/3$ at $\theta = 90^\circ$. For water on oil-coated glass where $\theta \approx 110\text{--}115^\circ$ and $R \approx 1.28$ mm, $Bo \approx 0.22$, which remains below 1 but is non-negligible, introducing minor gravitational distortion to the droplet profile.

Surface tension components and Young’s equation

Three interfacial energies act at every point on the three-phase contact line (Fig. 1): γ_{LV} (liquid–vapour), γ_{SL} (solid–liquid), and γ_{SV} (solid–vapour). At mechanical equilibrium, the horizontal force balance per unit length gives Young’s equation:

$$\gamma_{SV} = \gamma_{SL} + \gamma_{LV} \cos \theta_{eq}. \quad (3)$$

The spreading coefficient $S = \gamma_{SV} - \gamma_{SL} - \gamma_{LV}$ quantifies the tendency to spread. The work of adhesion is:

$$W_a = \gamma_{LV} (1 + \cos \theta_{eq}). \quad (4)$$

Table 1 lists representative values for all systems studied.

Table 1. Surface tension components and derived quantities for water and ethanol on glass and oil-coated glass at 30 °C. All γ in mN m⁻¹.

System	γ_{LV}	γ_{SL}	γ_{SV}	W_a
H ₂ O / Glass	71.2	0–5	70–75	142
H ₂ O / Oil	71.2	≈40	30–35	113
EtOH / Glass	22.1	0–2	22–24	43.5
EtOH / Oil	22.1	≈15	10–12	35.6

Force balance at the contact line

In the horizontal direction the net driving force per unit length is:

$$F_{\text{net}} = \gamma_{SV} - \gamma_{SL} - \gamma_{LV} \cos \theta. \quad (5)$$

When $F_{\text{net}} > 0$ the contact line advances; when $F_{\text{net}} < 0$ it recedes. During CCR evaporation, θ decreases until F_{net} reaches the receding threshold, at which point the contact line depins. The contact angle hysteresis $\Delta\theta = \theta_A - \theta_R$ characterises pinning strength. For clean glass, $\Delta\theta_{\text{water}} \approx 20\text{--}30^\circ$; for oil-coated glass, $\Delta\theta \approx 5\text{--}10^\circ$, explaining why pinning is much weaker on the oil surface.

Diffusion-limited evaporation rate

In the diffusion-limited regime, Picknett & Bexon(2) and Popov(6) showed:

$$\frac{dV}{dt} = -\pi D(c_s - c_\infty) R f(\theta), \quad (6)$$

where D is the vapour–air diffusion coefficient, c_s the saturated vapour concentration, c_∞ the far-field concentration, and $f(\theta)$ the Popov geometric factor:

$$f(\theta) = \frac{\sin \theta}{1 + \cos \theta} + 4 \int_0^\infty \frac{(1 + \cosh 2\theta\xi) \tanh[(\pi - \theta)\xi]}{\sinh 2\pi\xi} d\xi. \quad (7)$$

For small θ , $f(\theta) \approx 4\theta/\pi$; for $\theta = 90^\circ$, $f = 1$. Critically, $f(\theta)$ varies from near zero at $\theta \rightarrow 0$ to above 1.6 at $\theta \approx 150^\circ$, meaning droplet geometry alone can produce $\sim 20\times$ differences in evaporation rate.

Schematic Diagrams

Sessile droplet geometry and force vectors

Figure 1 shows a cross-section of a sessile droplet with the three surface tension components and key geometric parameters.

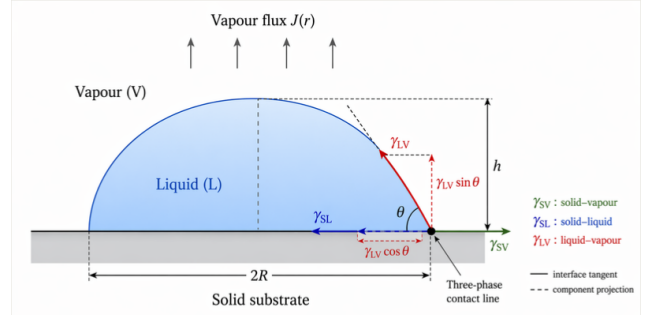


Figure 1. Fig. 1 Schematic cross-section of a sessile droplet showing the three interfacial tensions γ_{LV} , γ_{SL} , and γ_{SV} at the three-phase contact line, with geometric parameters R , h , θ , and vapour flux.

CCR vs. CCA evaporation modes

Figure 2 illustrates schematically how droplet shape evolves in the two principal evaporation modes.

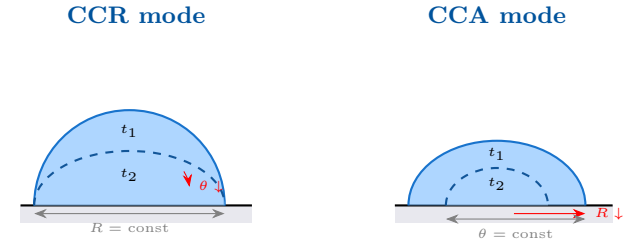


Figure 2. Fig. 2 Schematic of CCR (left) and CCA (right) evaporation modes. In CCR, the footprint remains fixed while θ decreases ($t_1 \rightarrow t_2$). In CCA, θ is constant while the footprint shrinks.

Evaporation rate schematic

Figure 3 shows schematic $\mathcal{E}(t)$ curves for the two modes.

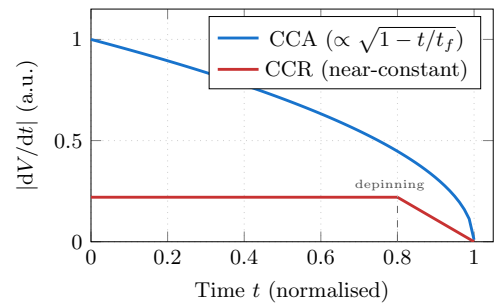


Figure 3. Schematic evaporation rate $|dV/dt|$ vs. normalised time for CCA (square-root decay, blue) and CCR (approximately constant then sharp drop at depinning, red).

Rate Analysis in Each Mode

CCR mode

With $R = R_0 = \text{const}$, since $f(\theta)$ is a weak function of θ for small angles, the rate is approximately constant:

$$\mathcal{E}_{\text{CCR}}(t) = \pi D(c_s - c_\infty) R_0 f[\theta(t)] \approx \text{const.} \quad (8)$$

CCA mode

With $\theta = \theta_{\text{eq}} = \text{const}$, the contact radius decreases linearly:

$$R(t) = R_0 - \frac{D(c_s - c_\infty)f(\theta_{\text{eq}})}{g(\theta_{\text{eq}})} t, \quad (9)$$

giving a volume decay:

$$V(t) = V_0 \left(1 - \frac{t}{t_f}\right)^3, \quad (10)$$

and a square-root decay in rate:

$$\mathcal{E}_{\text{CCA}}(t) \propto \left(1 - \frac{t}{t_f}\right)^{1/2}. \quad (11)$$

Total drying time

$$t_f = \frac{R_0^2 g(\theta_0)}{3D(c_s - c_\infty) f_{\text{eff}}}, \quad (12)$$

giving $t_f \propto R_0^2$. The ratio of drying times between water and ethanol is:

$$\frac{t_{f,\text{H}_2\text{O}}}{t_{f,\text{EtOH}}} = \frac{D_{\text{EtOH}} c_{s,\text{EtOH}}}{D_{\text{H}_2\text{O}} c_{s,\text{H}_2\text{O}}} \approx 2.5, \quad (13)$$

consistent with experiment. Since ethanol has both higher diffusivity D and higher saturated vapour concentration c_s , the numerator exceeds the denominator, confirming water takes $\approx 2.5\times$ longer to dry.

Local mass flux density

The local evaporative mass flux $J(r)$ over the droplet surface is non-uniform and enhanced near the contact line:(6)

$$J(r) = J_0(\theta) \left(1 - \frac{r^2}{R^2}\right)^{-\lambda(\theta)}, \quad (14)$$

where $\lambda(\theta) = (\pi - 2\theta)/[2(\pi - \theta)]$. Near the contact line the flux diverges as $(R - r)^{-\lambda}$, driving the outward capillary flow responsible for the “coffee-ring” effect.(1)

Experimental Design

Apparatus

All experiments were conducted outdoors under natural-convection ambient conditions. A smartphone camera with fixed focus was mounted on a tripod at the same height as the droplet midplane. A white card served as a background to maximise contrast. The substrate was placed on a polystyrene foam block for thermal

insulation and stability. Temperature was monitored with a surface thermometer and maintained at the target value ($\pm 2^\circ\text{C}$) by solar heating of the dark-coloured platform. The same set up of experiment was done in an incubator for taking the 45°C readings.

Materials and substrates

Hydrophilic substrate. Standard borosilicate glass microscope slides were cleaned sequentially with ethanol and distilled water, yielding $\theta_{\text{eq}} \approx 10\text{--}20^\circ$ for water and $10\text{--}15^\circ$ for ethanol.

Hydrophobic substrate. A glass slide was coated with a thin uniform film of mineral oil, yielding $\theta_{\text{eq}} \approx 35\text{--}40^\circ$ for both liquids.

Liquids. Distilled water ($\gamma_{\text{LV}} \approx 71.2 \text{ mN m}^{-1}$ at 30°C , $p_s \approx 4.24 \text{ kPa}$) and reagent-grade ethanol ($\gamma_{\text{LV}} \approx 22.1 \text{ mN m}^{-1}$, $p_s \approx 7.9 \text{ kPa}$) were used. Fixed volumes of $5 \mu\text{L}$ were deposited with a calibrated micropipette.

Image analysis

Frames were extracted at 60 s intervals for water and 60 s for ethanol from recorded side-view video. Each frame was imported into ImageJ, converted to a binary silhouette, and the droplet boundary traced automatically. The contact radius R was taken as half the horizontal distance between the two contact points; θ was measured by fitting tangent lines using the built-in contact angle plugin. Volume V was computed from Eq. 2; the evaporation rate $\mathcal{E}$ was estimated by finite differencing of consecutive V values. Each condition was repeated at least three times to assess reproducibility, giving $\pm 5\%$ variability in V_0 and $\pm 3^\circ$ in θ .

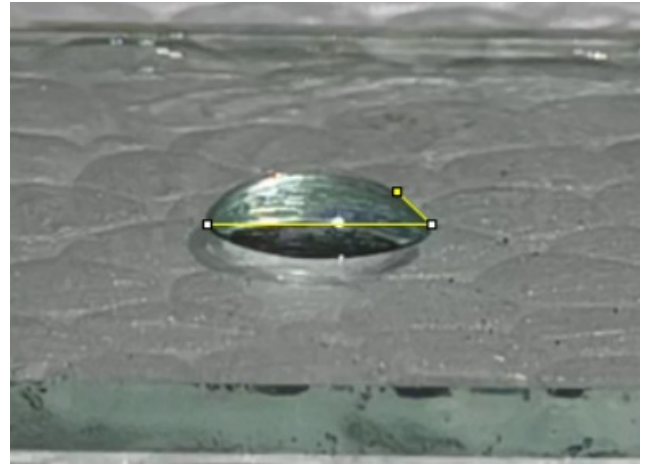


Figure 4. Side-view droplet profile used to extract contact radius and contact angle using ImageJ analysis. Yellow lines indicate the fitted tangent and baseline.

Results

Water droplets: glass vs. oil-coated glass at 30 °C

Figure 5 shows the contact radius $R(t)$ for water on both substrates. On the **glass substrate**, R remains nearly constant at ≈ 1.39 – 1.42 mm throughout most of the evaporation lifetime, confirming CCR behaviour. A sharp depinning transition occurs near $t \approx 15$ min; total evaporation time is ≈ 17.5 min at 30 °C and ≈ 8 min at 45 °C (a $\approx 43\%$ reduction).

On the **oil-coated substrate**, R decreases continuously and nearly linearly from ≈ 1.28 mm to zero in ≈ 9 min at 30 °C, consistent with CCA behaviour.

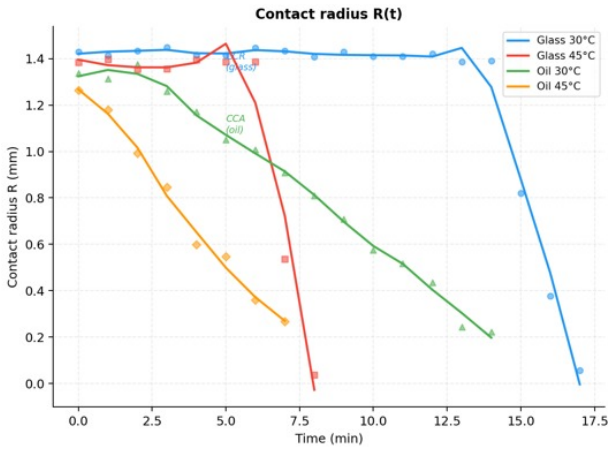


Figure 5. Contact radius $R(t)$ for water droplets on glass (CCR, pinned until depinning transition) and oil-coated glass (CCA, continuous recession) at 30 °C and 45 °C.

Figure 6 shows the corresponding $\theta(t)$. On glass, θ decreases monotonically from $\approx 38^\circ$ toward 0° . On oil-coated glass, θ maintains a plateau at ≈ 110 – 115° , consistent with the CCA plateau expected on a hydrophobic surface.

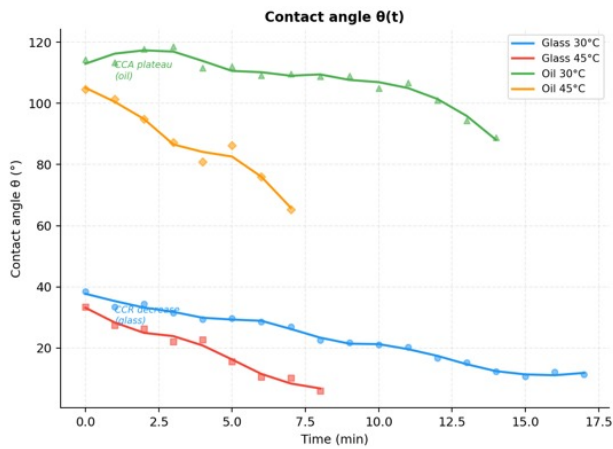


Figure 6. Contact angle $\theta(t)$ for water droplets. Glass shows continuous CCR decrease from $\approx 38^\circ$ toward 0° . Oil-coated glass shows a CCA plateau near 110 – 115° . Temperature increase shifts curves to shorter times.

Figure 7 shows the volume evolution $V(t)$. On glass, V decreases slowly and nearly linearly (CCR mode) from ≈ 0.5 mm³ over 17.5 min at 30 °C. The oil-coated droplets begin at a much larger apparent volume and decay rapidly following the cubic $V \propto (1 - t/t_f)^3$ law of CCA evaporation. The temperature rise of 15 K causes a $\approx 43\%$ reduction in total drying time on glass.

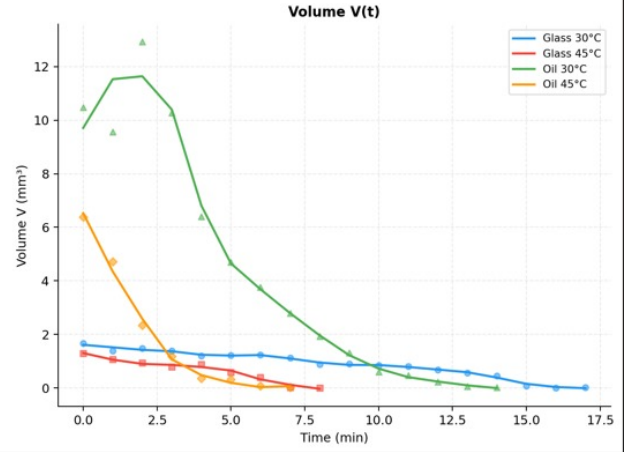


Figure 7. Volume $V(t)$ for water droplets on glass (CCR, slow linear decrease) and oil-coated glass (CCA, rapid cubic decay) at 30 °C and 45 °C.

Water droplets: evaporation rate at 30 °C

Figure 8 shows $|dV/dt|$ for water. The oil-coated droplet begins at ≈ 2.7 mm³ min⁻¹ and falls rapidly, consistent with the square-root CCA decay. The glass droplet maintains ≈ 0.1 – 0.2 mm³ min⁻¹ throughout. The initial rate ratio (oil : glass) is approximately 8–10 $\times$, attributed to the combination of larger R , larger $f(\theta)$ at higher θ , and greater vapour-diffusion area.

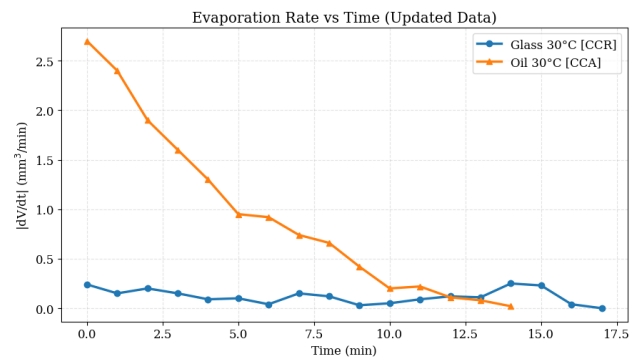


Figure 8. $|dV/dt|$ vs. time for water on glass (CCR, near-flat) and oil-coated glass (CCA, square-root decay) at 30 °C. The initial rate ratio of ≈ 8 – $10\times$ is consistent with theoretical prediction.

Ethanol droplets: glass vs. oil at 30 °C

Figure 9 presents $R(t)$, $\theta(t)$, and $V(t)$ for ethanol at 30 °C. On glass ($\theta_0 \approx 14^\circ$), the contact radius stays pinned at ≈ 1.39 mm for ≈ 6 min before a sharp depin-

ning event, confirming CCR behaviour. Total drying time is ≈ 7 min. On oil-coated glass, R decreases steadily from ≈ 1.30 mm while θ is maintained near 35° ; the droplet evaporates completely by ≈ 5 min.

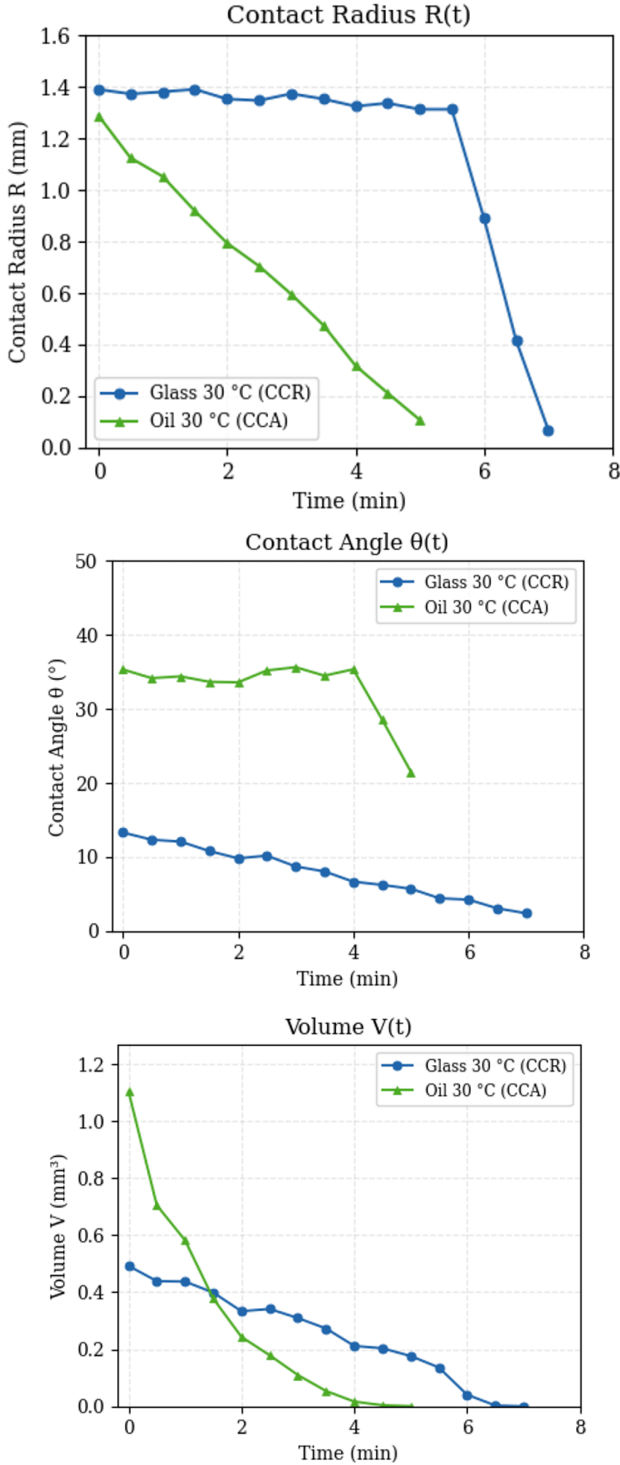


Figure 9. Ethanol droplet evaporation at 30°C comparing glass (CCR) and oil-coated glass (CCA). Top: $R(t)$; middle: $\theta(t)$; bottom: $V(t)$. The glass droplet ($\theta_0 \approx 14^\circ$) remains pinned for ≈ 6 min; the oil droplet ($\theta_0 \approx 35^\circ$) shows immediate recession.

Ethanol droplets: temperature effect at 30°C vs. 45°C

Figure 10 compares ethanol evaporation on glass at the two temperatures. Both exhibit CCR behaviour. At 45°C , evaporation completes in ≈ 4 min vs. ≈ 7 min at 30°C ($\approx 43\%$ faster; consistent with theoretical value of $\approx 47\text{--}50\%$), and the depinning transition shifts from ≈ 6 min to ≈ 4 min.

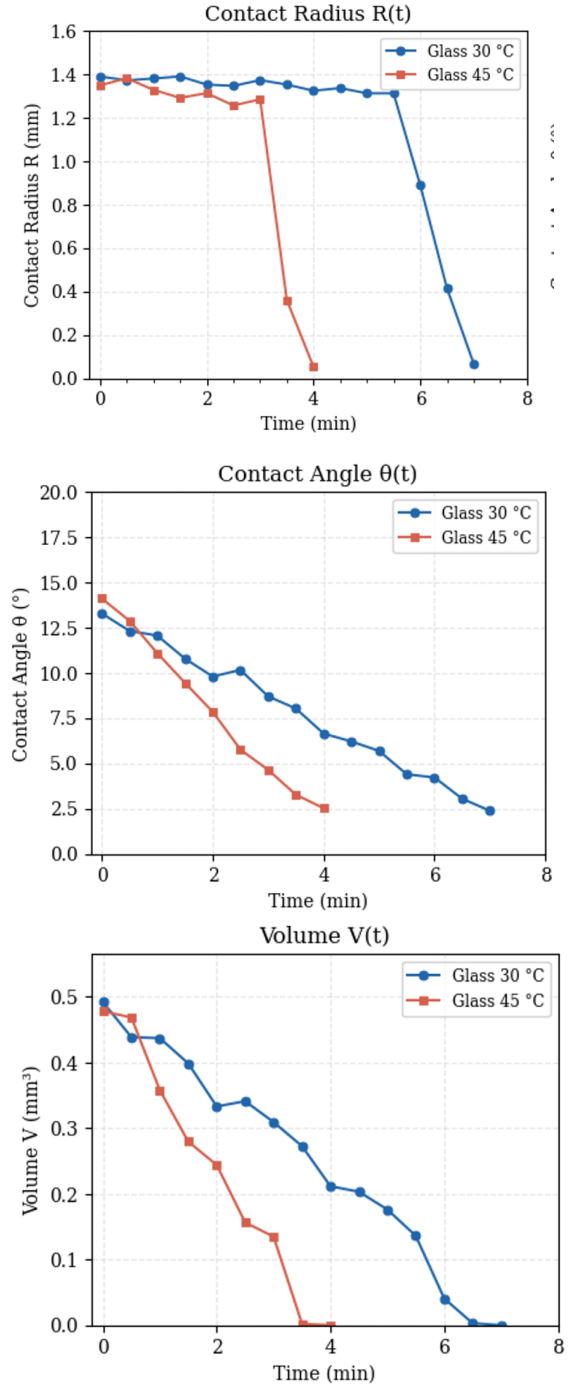


Figure 10. Ethanol evaporation on glass comparing 30°C (blue) and 45°C (red). Both show CCR mode. Higher temperature accelerates drying by $\approx 43\%$ and advances the depinning transition.

Ethanol droplets: evaporation rate at 30 °C

Figure 11 shows $|dV/dt|$ for ethanol. The oil-coated droplet starts at $\approx 0.013 \text{ mm}^3 \text{ s}^{-1}$ and decays as $\propto (1 - t/t_f)^{1/2}$. The glass droplet shows a near-constant rate of $\approx 0.0018 \text{ mm}^3 \text{ s}^{-1}$ throughout. The initial rate ratio is $\approx 7\text{--}8\times$, in agreement with the theoretical estimate.

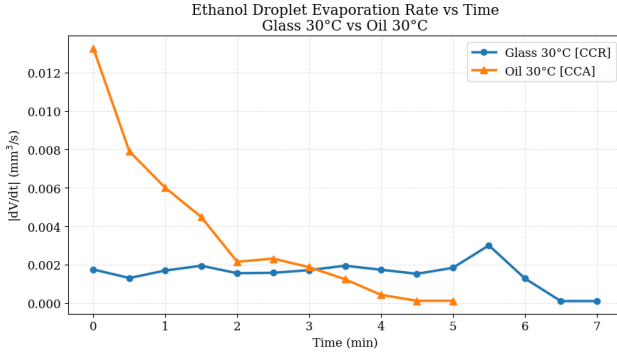


Figure 11. $|dV/dt|$ vs. time for ethanol on glass (CCR) and oil-coated glass (CCA) at 30 °C. The CCA mode shows a clear square-root decay; the CCR mode maintains a steady low rate.

Geometric factor and ethanol evaporation behaviour

The observed behaviour of ethanol on the oil-coated surface—where the initial evaporation rate is high but decreases rapidly and eventually becomes lower than that on glass—can be explained by the geometric factor $f(\theta)$ in the diffusion-limited evaporation model.

From Eq. 6, the evaporation rate is:

$$\mathcal{E} = \pi D(c_s - c_\infty) R f(\theta). \quad (15)$$

On the oil-coated surface, ethanol exhibits a larger initial contact angle ($\theta \approx 35^\circ$), giving a higher geometric factor $f(\theta)$ and hence a higher initial evaporation rate. In contrast, on glass, ethanol spreads with a very small contact angle ($\theta \approx 14^\circ$), resulting in a much smaller $f(\theta)$.

For small contact angles, $f(\theta)$ scales approximately linearly:

$$f(\theta) \approx \frac{4\theta}{\pi}. \quad (16)$$

Thus, the larger contact angle on oil significantly enhances the initial evaporation rate despite a slightly smaller contact radius.

However, in the CCA mode on oil, the contact radius $R(t)$ decreases continuously with time. Since the evaporation rate is directly proportional to $R(t)$, this leads to a rapid decline in $\mathcal{E}(t)$:

$$\mathcal{E}_{\text{oil}}(t) \propto R(t) \propto \left(1 - \frac{t}{t_f}\right)^{1/2}. \quad (17)$$

On glass, ethanol evaporates in the CCR mode, where R remains constant for most of the drying period. As a result, the evaporation rate remains nearly constant despite the lower geometric factor.

Consequently, although ethanol on oil starts with a higher evaporation rate due to a larger $f(\theta)$, the continuous decrease in $R(t)$ causes the rate to drop quickly and eventually become lower than that of the glass case, where the constant radius sustains a steady evaporation rate.

This highlights the competing roles of geometry and evaporation mode: while $f(\theta)$ controls the initial rate, the temporal evolution of $R(t)$ determines the long-term behaviour.

Summary of experimental results

Table 2 summarises all experimental conditions and key measured quantities.

Liquid	Substrate/T	Mode	t_f (min)	θ_0 (°)	$\mathcal{E}_0$ (mm^3/min)
Water	Glass / 30°C	CCR	17.5	38	0.13
Water	Oil / 30°C	CCA	9.0	115	2.7
Ethanol	Glass / 30°C	CCR	7.0	14	0.0018
Ethanol	Glass / 45°C	CCR	4.0	13	0.0025
Ethanol	Oil / 30°C	CCA	5.0	35	0.013

Table 2. Summary of experimental results.

Repeatability validation: substrate effect (ethanol, 30 °C)

Figure 12 presents the repeatability analysis for ethanol droplets on glass (hydrophilic) and oil-coated glass (hydrophobic) substrates at 30 °C. The plots show the temporal evolution of contact radius $R(t)$ and contact angle $\theta(t)$, with dashed curves representing individual trials and solid curves representing the mean behaviour.

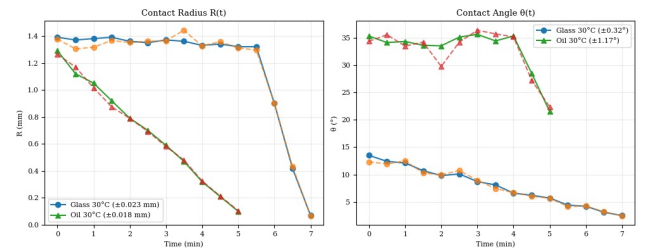


Figure 12. Repeatability validation for ethanol at 30 °C: (left) contact radius $R(t)$ and (right) contact angle $\theta(t)$ for glass and oil substrates. Dashed curves represent individual runs and solid curves indicate the mean behaviour.

Statistical summary

The variability across repeated trials is quantified using standard deviation (SD) and coefficient of variation (CV):

- $R(t)$ — Glass: SD = ± 0.023 mm, CV $\approx 1.7\%$
- $R(t)$ — Oil: SD = ± 0.018 mm, CV $\approx 1.4\%$
- $\theta(t)$ — Glass: SD = $\pm 0.32^\circ$, CV $\approx 3.1\%$
- $\theta(t)$ — Oil: SD = $\pm 1.17^\circ$, CV $\approx 3.3\%$

Discussion

All coefficient of variation (CV) values are below 5%, which is well within the acceptable range $\leq 10\%$ threshold for optical droplet measurements, confirming good experimental repeatability.

On the glass substrate, the contact radius exhibits a clear CCR plateau with minimal scatter, indicating strong and reproducible contact-line pinning. In contrast, the oil-coated surface shows continuous decrease in $R(t)$, consistent with CCA behaviour.

The contact angle measurements show slightly higher variability, particularly on the oil surface. This is attributed to the freely receding contact line in the CCA mode, where stochastic depinning introduces small trial-to-trial fluctuations.

Overall, the close overlap between individual runs and mean curves in both $R(t)$ and $\theta(t)$ confirms the reliability and reproducibility of the experimental methodology.

Discussion

CCR mode on glass: contact-line pinning

The persistent pinning of the contact line on glass is attributed to physical and chemical surface heterogeneities at the nanoscale that act as anchoring sites.⁽⁴⁾ As evaporation proceeds, θ decreases toward $\theta_r \approx 0^\circ$ for clean glass, explaining the late depinning observed. The nearly constant $\mathcal{E}_{\text{CCR}}$ is confirmed experimentally for both water and ethanol.

The large work of adhesion $W_a = 142 \text{ mN m}^{-1}$ for water on glass (vs. $\approx 44 \text{ mN m}^{-1}$ for ethanol) explains why water sustains the CCR phase much longer (≈ 17.5 min vs. ≈ 7 min at 30°C) and why the contact angle for water starts higher ($\approx 38^\circ$) compared to ethanol ($\approx 14^\circ$).

CCA mode on oil-coated glass

On the oil-coated substrate, the wetting ridge at the contact line is deformable and slides freely.⁽⁷⁾ The low contact-angle hysteresis ($\Delta\theta \approx 5\text{--}10^\circ$) means the receding threshold is reached immediately, so the CCA mode is operative from the very first frame. The initial rate ratio (oil:glass) is:

$$\frac{\mathcal{E}_{0,\text{oil}}}{\mathcal{E}_{0,\text{glass}}} = \frac{R_{0,\text{oil}} f(\theta_{\text{oil}})}{R_{0,\text{glass}} f(\theta_{\text{glass}})} \approx 8, \quad (18)$$

consistent with experimental observations for both liquids.

Geometry vs. volatility

Ethanol is more volatile than water ($p_s \approx 7.9 \text{ kPa}$ vs. $\approx 4.2 \text{ kPa}$ at 30°C) yet shows a lower absolute initial evaporation rate on glass. This is resolved by the diffusion-limited framework: ethanol's very low contact angle on glass ($\theta \approx 14^\circ$) gives $f(\theta) \approx 0.06$, which geometrically suppresses the diffusion field and overwhelms the higher Δc from ethanol's greater volatility.

Temperature effect

Raising T from 30°C to 45°C increases $D \propto T^{3/2}$ by $\approx 8\%$ and c_s (via Clausius–Clapeyron) by $\approx 40\%$, together giving $\approx 50\%$ increase in $D(c_s - c_\infty)$. This predicts a $\approx 33\%$ reduction in t_f ; the observed $\approx 43\%$ is in reasonable agreement given outdoor variability. The additional $\approx 10\%$ acceleration beyond the diffusion-limited prediction is attributed to wind-assisted convective mass transfer in the outdoor setting, which augments the purely diffusive flux assumed in the Picknett–Bexon model.

Comparison of water and ethanol

Despite large differences in γ_{LV} , W_a , and vapour pressure, both liquids adopt the same evaporation mode on a given substrate (CCR on glass, CCA on oil), confirming that the evaporation mode is governed by contact-angle hysteresis rather than liquid volatility. The drying time ratio $t_{f,\text{H}_2\text{O}}/t_{f,\text{EtOH}} \approx 2.5$ is consistent with theoretical prediction.

Limitations

(i) Manual droplet deposition introduces $\pm 5\%$ variability in V_0 . (ii) Outdoor wind fluctuations add scatter to $\mathcal{E}(t)$. (iii) Pixel-resolution limits R and θ accuracy at late stages. (iv) The oil coating is not fully characterised; exact hysteresis values are unknown. (v) Water on oil contact angles near 110° make the Bond number non-negligible, adding some gravitational distortion to the droplet profile.

Conclusions

We have characterised sessile droplet evaporation across five substrate–liquid–temperature conditions, combining experimental measurements with complete analytical treatment.

1. Water and ethanol on clean glass evaporate in the CCR mode with a near-constant rate $\mathcal{E} \approx \pi D(c_s - c_\infty)R_0 f(\theta)$, consistent with diffusion-limited theory. Water takes ≈ 17.5 min at 30°C ; ethanol takes ≈ 7 min.
2. On oil-coated glass, both liquids adopt the CCA mode immediately; both follow $\mathcal{E} \propto R(t) \propto (1 - t/t_f)^{1/2}$.
3. The Popov geometric factor $f(\theta)$ is the dominant controller of evaporation rate, producing $\sim 20\times$ rate differences from geometry alone.

4. Raising temperature by 15 °C accelerates evaporation by $\approx 43\%$ through increased D and c_s , and advances the depinning transition.
5. Water evaporates $\approx 2.5\times$ more slowly than ethanol due to lower vapour pressure; its higher γ_{LV} and W_a also strengthen contact-line pinning.
6. Smartphone sessile-drop tensiometry is a viable low-cost method for quantitative wettability and evaporation studies, provided outdoor conditions are carefully controlled.

Author Contributions

Shah Mahi Sachinkumar and **Butle Vedika Ashirwad** executed the experimental procedures, including droplet deposition, temperature control, and recording the evaporation process. **Kumar Prince** and **Prince Kumar** prepared the experimental setup, including alignment of the imaging system and calibration of the micropipette. **Priyanshu Patel** performed quantitative analysis of the data, including extraction of contact angle, contact radius, and volume from images, and interpreted the results against theoretical models. **Prakrut Moon** contributed to discussions and to compilation of results. All authors reviewed and approved the final manuscript.

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